

ECEN 662: Estimation & Detection Theory

Lecture 8: The Expectation-Maximization Algorithm

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Models with latent variables

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- The above optimization problem can be difficult to solve analytically due to the *marginalization* of the latent variables, which occurs *inside* the logarithm of the objective function
- Next, let us use the *mixture Gaussian model* as a concrete example to see what exactly the aforementioned challenge is and how the problem may be solved with *iterative numerical techniques*

Mixture Gaussian model

- Let $X = (X_1, \dots, X_n)$ and $U = (U_1, \dots, U_n)$, where (X_i, U_i) are i.i.d. according to:

$$U_i \sim \text{Cat}(u_i; \pi_1, \dots, \pi_K) \quad \text{and} \quad X_i | U_i \sim \mathcal{N}(x_i; \mu_{u_i}, \sigma_{u_i}^2)$$

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- Marginalizing* over U_i gives

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which is a *mixture Gaussian* distribution, and

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- The log-likelihood for all measurements is given by

$$\log f_{\theta}(x) = \sum_{i=1}^n \log f_{\theta}(x_i) = \sum_{i=1}^n \log \left(\sum_{k=1}^K \pi_k \mathcal{N}(x_i; \mu_k, \sigma_k^2) \right)$$

Finding MLE for $(\mu_1, \dots, \mu_K)$

- Assume for the moment that $(\pi_1, \dots, \pi_K)$ and $(\sigma_1^2, \dots, \sigma_K^2)$ are known. To find the MLE for $(\mu_1, \dots, \mu_K)$, note that for any $j \in [K]$ we have

$$\frac{\partial}{\partial \mu_j} \log f_\theta(x) = \sum_{i=1}^n \frac{\tilde{\pi}_j(x_i)(x_i - \mu_j)}{\sigma_j^2}$$

where

$$\tilde{\pi}_j(x_i) = \frac{\pi_j \mathcal{N}(x_i; \mu_j, \sigma_j^2)}{\sum_{k=1}^K \pi_k \mathcal{N}(x_i; \mu_k, \sigma_k^2)} = \mathbb{P}_\theta(U_i = j | X_i = x_i)$$

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- Setting the partial derivative equal to zero for all j gives the following set of equations:

$$\sum_{i=1}^n \frac{\tilde{\pi}_j(x_i)(x_i - \mu_j)}{\sigma_j^2} = 0, \quad \forall j \in [K]$$

for which the solution for $(\mu_1, \dots, \mu_K)$ is the MLE

A vicious circle

- Note that if $\tilde{\pi}_j(x_i)$'s are *known*, the above equations are *separate* for each μ_j and hence are very easy to solve:

$$\mu_j = \frac{\sum_{i=1}^n \tilde{\pi}_j(x_i) x_i}{\sum_{i=1}^n \tilde{\pi}_j(x_i)}, \quad \forall j \in [K]$$

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- One man's vicious circle is another man's *successive approximation* procedure

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 - 4) Stop if the value for $(\mu_1, \dots, \mu_K)$ has already *converged*; otherwise go back to Step 2
- Next, we shall *generalize* the above procedure to general likelihood models with *latent* variables, and the resulting procedure is known as the *expectation-maximization (EM)* algorithm

A variational characterization of the log-likelihood

- The main idea behind the *EM* algorithm is the following *variational* characterization of the log-likelihood:

$$\log f_{\theta}(x) = \max_q F(q, \theta)$$

where

$$F(q, \theta) = \log f_{\theta}(x) - D_{KL}(q(u) \| f_{\theta}(u|x)) \quad (1)$$

$$= \mathbb{E}_q [\log f_{\theta}(x, U)] - \mathbb{E}_q [\log q(U)], \quad (2)$$

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$$(\hat{\theta}, q^*) \in \arg \max_{(\theta, q)} F(q, \theta)$$

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- q^* can be disregarded if our sole objective is to find the MLE for θ , but is usually quite useful under specific settings

- First note that by the *information* inequality, for any density function q we have

$$D_{KL}(q(u) \| f_{\theta}(u|x)) \geq 0$$

where the equality holds if and only if

$$q(u) = f_{\theta}(u|x), \quad \forall u \in \mathcal{U}$$

Proof

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- We thus have

$$\log f_{\theta}(x) \geq \log f_{\theta}(x) - D_{KL}(q(u) \| f_{\theta}(u|x))$$

where the equality holds if and only if

$$q(u) = f_{\theta}(u|x), \quad \forall u \in \mathcal{U}$$

and hence

$$\log f_{\theta}(x) = \max_q (\log f_{\theta}(x) - D_{KL}(q(u) \| f_{\theta}(u|x)))$$

- Further note that

$$\begin{aligned}\log f_{\theta}(x) - D_{KL}(q(u) || f_{\theta}(u|x)) \\&= \log f_{\theta}(x) - (\mathbb{E}_q [\log q(U)] - \mathbb{E}_q [\log f_{\theta}(U|x)]) \\&= \mathbb{E}_q [\log f_{\theta}(x)] + \mathbb{E}_q [\log f_{\theta}(U|x)] - \mathbb{E}_q [\log q(U)] \\&= \mathbb{E}_q [\log (f_{\theta}(x)f_{\theta}(U|x))] - \mathbb{E}_q [\log q(U)] \\&= \mathbb{E}_q [\log f_{\theta}(x, U)] - \mathbb{E}_q [\log q(U)]\end{aligned}$$

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- We have thus proved that expressions (1) and (2) are in fact *equivalent* to each other

Finding MLE via alternating maximization

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- Consider the following *alternating-maximization* strategy for solving the above *double-maximization* problem:
 - Initialize $\theta = \theta^{(0)}$ and $t = 1$
 - Repeat the following steps until the values of (θ, q) converge:
 - Fix $\theta = \theta^{(t-1)}$ and find
$$q^{(t)} \in \arg \max_q F(q, \theta^{(t-1)})$$
 - Fix $q = q^{(t)}$ and find
$$\theta^{(t)} = \arg \max_{\theta} F(q^{(t)}, \theta)$$
 - $t \leftarrow t + 1$

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- By expression (1) and the information inequality,

$$q^{(t)}(u) = f_{\theta^{(t-1)}}(u|x), \quad \forall u \in \mathcal{U}$$

which gives

$$F \left(q^{(t)}, \theta^{(t-1)} \right) = \log f_{\theta^{(t-1)}}(x)$$

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- By comparison, the problem of fixing $q = q^{(t)}$ and finding

$$\theta^{(t)} \in \arg \max_{\theta} F(q^{(t)}, \theta)$$

is much more involved

- To solve this optimization problem, we shall use expression (2) and write

$$F\left(q^{(t)}, \theta\right)=\mathbb{E}_{q^{(t)}}\left[\log f_{\theta}(x, U)\right]-\mathbb{E}_{q^{(t)}}\left[\log q^{(t)}(U)\right]$$

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- Thus, $\theta^{(t)}$ can be found by maximizing over the first term in the above expression:

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- The above two equations allow us to continuously update the value of (θ, q) , leading to the famous *EM* algorithm

The EM algorithm

- The algorithm:
 - Initialize $\theta = \theta^{(0)}$ and $t = 1$
 - Do the following steps until the values of (θ, q) converge:
 - ▶ (E-step) Compute the *expectation*

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- Note that

$$\begin{aligned} \log f_{\theta^{(t-1)}}(x) &= F(q^{(t)}, \theta^{(t-1)}) \\ &\leq F(q^{(t)}, \theta^{(t)}) \leq F(q^{(t+1)}, \theta^{(t)}) = \log f_{\theta^{(t)}}(x) \end{aligned}$$

so the algorithm increases *monotonically* the log-likelihood $\log f_{\theta}(x)$

Back to the mixture Gaussian model

- Recall that the M-step in each iteration is to solve

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- For the *mixture Gaussian* model, we have

$$\begin{aligned} & \mathbb{E}_{q^{(t)}} [\log f_{\theta}(x, U)] \\ &= \sum_{i=1}^n \mathbb{E}_{q^{(t)}} [\log f_{\theta}(x_i, U_i)] \\ &= \sum_{i=1}^n \mathbb{E}_{q^{(t)}} [\log f_{\theta}(x_i|U_i)] + \sum_{i=1}^n \mathbb{E}_{q^{(t)}} [\log f_{\theta}(U_i)] \\ &= \sum_{i=1}^n \sum_{k=1}^K \pi_k^{(t)}(x_i) \log \mathcal{N}(x_i; \mu_k, \sigma_k^2) + \sum_{i=1}^n \sum_{k=1}^K \pi_k^{(t)}(x_i) \log \pi_k \end{aligned}$$

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- These estimates reduce to the standard empirical mean, variance, and frequency estimates when the posterior probability vectors $(\pi_1^{(t)}(x_i), \dots, \pi_K^{(t)}(x_i))$ are *0-1* vectors

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- *Variational EM (VEM)*: Approximates the posterior distribution in the E-step using a *simple, tractable* distribution
- *Expectation conditional maximization (ECM)*: Performs the M-step by updating parameters *one at a time*
- *Generalized EM (GEM)*: Instead of performing the M-step exactly, it only requires a *non-decreasing* update